

Aim

To analyze the Car Prediction dataset, perform data cleaning and data transformation, create a derived Car_Age feature, perform exploratory analysis, build a Random Forest Regression model to predict Selling_Price, evaluate the model, and export the processed data to Excel.

Procedure

1. Load the Car Prediction dataset using Pandas.
2. Display the first few records, shape, information, and missing values.
3. Check and handle missing values.
4. Create Car_Age from the Year column.
5. Perform descriptive statistics and exploratory analysis.
6. Analyze selling price by Fuel Type, Seller Type, and Transmission.
7. Prepare numerical and categorical features for machine learning.
8. Split the dataset into training and testing data.
9. Train a Random Forest Regression model.
10. Predict selling price for the test dataset.
11. Evaluate the model using MAE, RMSE, and R^2 .
12. Export the cleaned and transformed dataset to Excel.

Python Code

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.compose import ColumnTransformer
```

```

from sklearn.preprocessing import OneHotEncoder
from sklearn.pipeline import Pipeline
from sklearn.impute import SimpleImputer
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
df = pd.read_csv("car_prediction_data.csv")
print(df.head())
print(df.shape)
print(df.info())
print(df.isnull().sum())
df["Car_Age"] = 2026 - df["Year"]
print(df.describe())
print(df.groupby("Fuel_Type")["Selling_Price"].mean())
print(df.groupby("Seller_Type")["Selling_Price"].mean())
print(df.groupby("Transmission")["Selling_Price"].mean())
df.groupby("Fuel_Type")["Selling_Price"].mean().plot(kind="bar")
plt.title("Average Selling Price by Fuel Type")
plt.show()
df.groupby("Seller_Type")["Selling_Price"].mean().plot(kind="bar")
plt.title("Average Selling Price by Seller Type")
plt.show()
X = df.drop(columns=["Selling_Price"])
y = df["Selling_Price"]

num_cols = X.select_dtypes(include=np.number).columns.tolist()
cat_cols = X.select_dtypes(exclude=np.number).columns.tolist()

preprocessor = ColumnTransformer([
    ("num", SimpleImputer(strategy="median"), num_cols),
    ("cat", Pipeline([
        ("imputer", SimpleImputer(strategy="most_frequent")),
        ("onehot", OneHotEncoder(handle_unknown="ignore"))
    ]), cat_cols)])
model = Pipeline([
    ("preprocessor", preprocessor),
    ("regressor", RandomForestRegressor(n_estimators=200, random_state=42))]
X_train, X_test, y_train, y_test = train_test_split(

```

```
X, y, test_size=0.2, random_state=42)
```

```
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
```

```
mae = mean_absolute_error(y_test, y_pred)
rmse = np.sqrt(mean_squared_error(y_test, y_pred))
r2 = r2_score(y_test, y_pred)
```

```
print("MAE:", mae)
print("RMSE:", rmse)
print("R2 Score:", r2)
```

```
df.to_excel("Car_Prediction_Cleaned_Transformed.xlsx", index=False)
```

Output

First five records:

Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Transmission	Owner
ritz	2014	3.35	5.59	27000	Petrol	Dealer	Manual	0
sx4	2013	4.75	9.54	43000	Diesel	Dealer	Manual	0
ciaz	2017	7.25	9.85	6900	Petrol	Dealer	Manual	0
wagon r	2011	2.85	4.15	5200	Petrol	Dealer	Manual	0
swift	2014	4.6	6.87	42450	Diesel	Dealer	Manual	0

Dataset Shape: (301, 9)

Missing Values: No missing values were found in the supplied dataset.

Average Selling Price: 4.6613

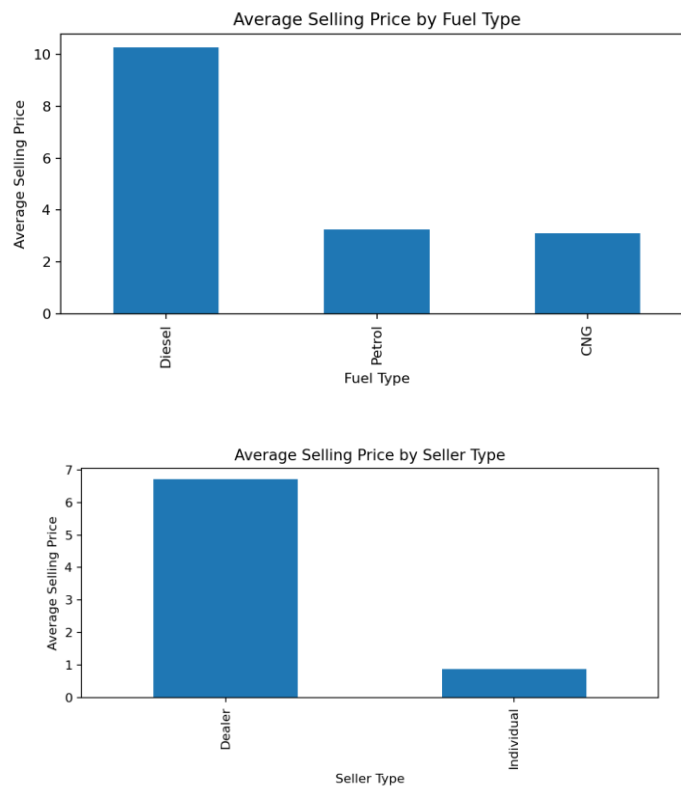
Minimum Selling Price: 0.1000

Maximum Selling Price: 35.0000

EDA Results

Highest average selling price by Fuel Type: Diesel (10.2785)

Highest average selling price by Seller Type: Dealer (6.7217)



Model Evaluation

Training records: 240

Testing records: 61

MAE: 0.5901

RMSE: 0.8872

R² Score: 0.9658

Result

The Car Prediction dataset was successfully cleaned and transformed. A Car_Age feature was derived from Year, numerical and categorical variables were prepared, and exploratory analysis was performed. A Random Forest Regression model achieved an R² score of 0.9658 on the test data. The processed dataset was exported to Car_Prediction_Cleaned_Transformed.xlsx.